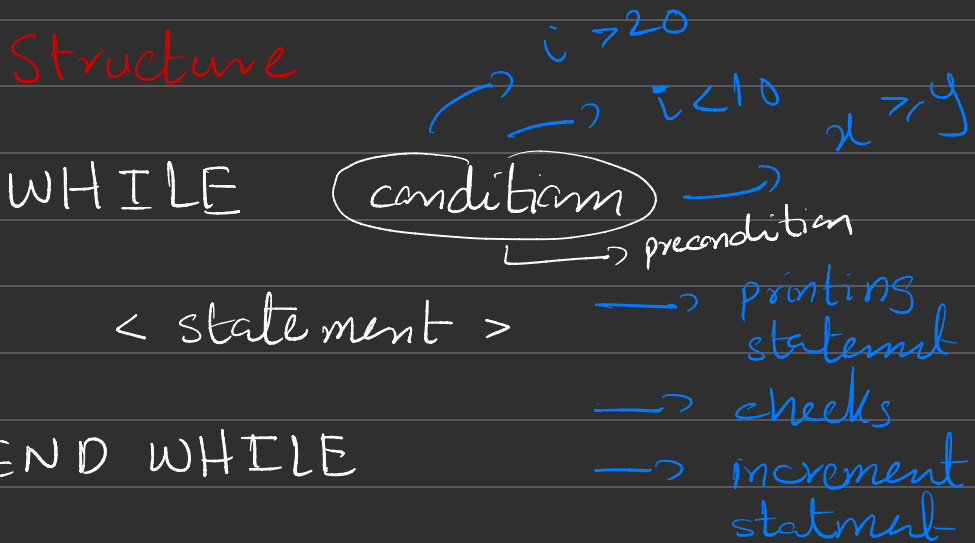


WHILE LOOPS

- specially used when you need to ^{no. of} can't
- used when the iterations are not well defined.
- used to perform a kind check before any / every iteration.

Structure



END WHILE

When to use:

- ① validating input from user
- ② looping until a sensor trigger.



Example

Q Print first 10 numbers using while

```
① DECLARE count ← 0 1
② WHILE count ≤ 10. 2
③   OUTPUT count      3
④   count ← count + 1 4 5 6 7 8 9 10
⑤ END WHILE           9
```

REPEAT UNTIL

A post check loop that runs atleast once:

Syntax

REPEAT

<statement>

UNTIL (condition) — post condition

When to use

- ① Asking user input atleast once
- ② Ensuring minimum one iterations.

Examples

REPEAT

INPUT Password

until password = "exit"

statement

condition

Q Display number 1 to 10

REPEAT

OUTPUT counter

counter $\leftarrow$ counter + 1

until counter = 10

Loop Types

condition chk

Run At least
once

Best when

FOR

Pre Condition

X

You know iteration count

WHILE

Pre condition

X

condition iteration, unknown

REPEAT
UNTIL

Post Condition

✓

Runs at least
once